



Republic of Iraq

Middle Technical University

Technical Engineering College of Artificial Intelligence

Cybersecurity Engineering Technology Department

Computer Network Protocols

Chapter Two

The Internet Protocol (IP) **(IPv4 – IPv6)**

Prepared By:

Assistant Prof. Dr. Mahmoud Shuker Mahmoud

mahmoud.shukur@mtu.edu.iq

2.1- Introduction

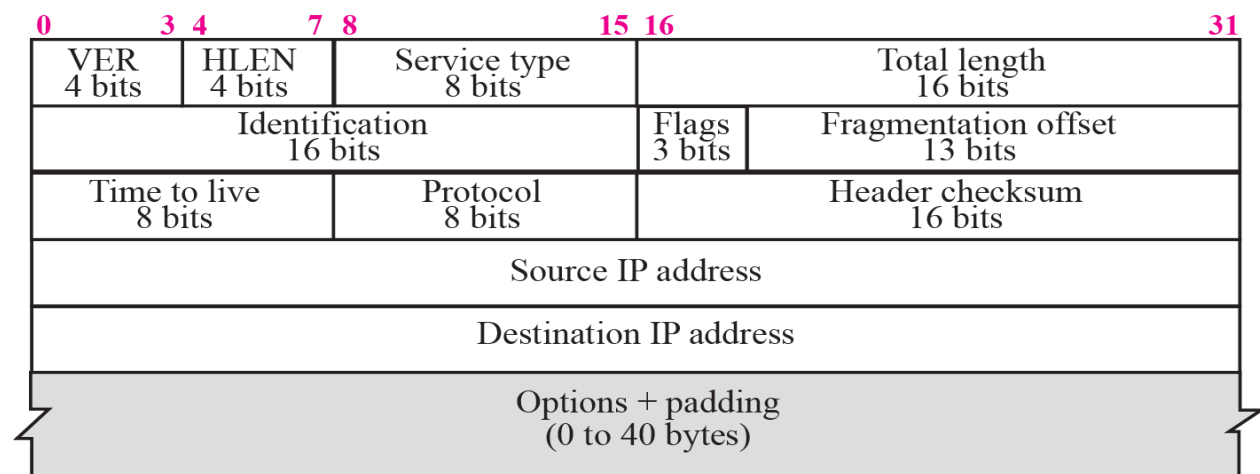
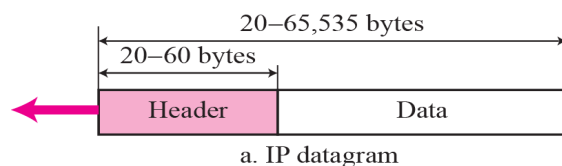
Communication at the network layer is source to destination; a computer somewhere in the world needs to communicate with another computer somewhere else in the world. Usually, computers communicate through the Internet. The packet transmitted by the sending computer may pass through several LANs or WANs before reaching the destination computer.

2.2- IPv4 Addressing

In the most widely installed level of the Internet Protocol (IP) today, an IPv4 (IP version 4) addresses or simply IP addresses if there is no confusion. It is a **32-bit address** that *uniquely* and *universally* defines the connection of a device (for example, a computer or a router) to the Internet.

2.3- IPv4 Datagram Format

Packets in the network (internet) layer are called **datagrams**. A datagram is a variable-length packet consisting of two parts: *header and data*. The header is **20 to 60 bytes** in length and contains information essential to routing and delivery. It is customary in TCP/IP to show the header in 4-byte sections.



b. Header format

IPv4 Header Structure

The IPv4 header typically contains 14 fields, 13 of which are mandatory.

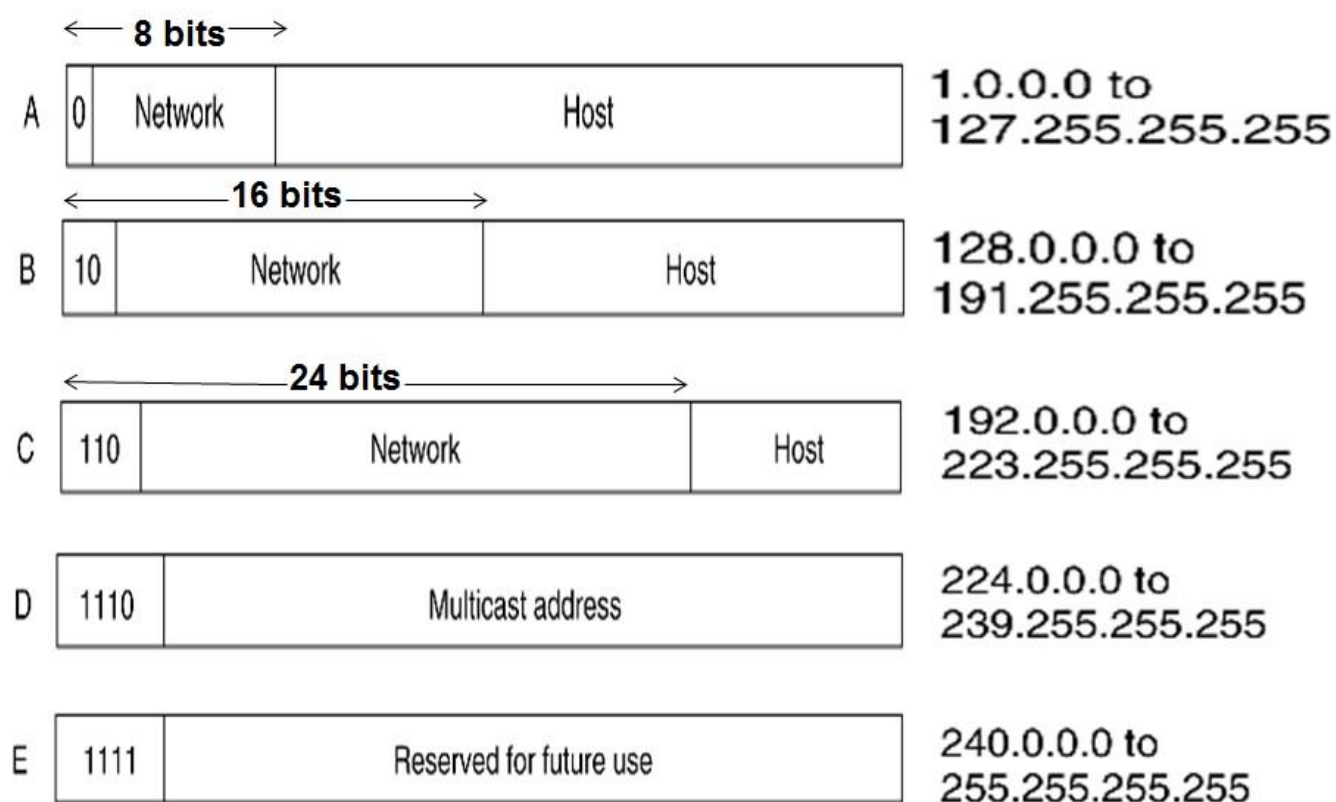
- **Version (4 bits):** Identifies the IP version. For IPv4, this is always 0100 (decimal 4).
- **Header Length (IHL) (4 bits):** Specifies the header size in 32-bit words. A value of 5 indicates the minimum 20-byte header, while 15 indicates the maximum 60-byte header.
- **Type of Service (ToS) / DSCP (8 bits):** Indicates the priority and quality of service (QoS) requirements for the datagram.
- **Total Length (16 bits):** Defines the entire size of the datagram (header + data) in bytes. The maximum possible size is 65,535 bytes.
- **Identification (16 bits):** Used to uniquely identify fragments of an original IP datagram to help with reassembly.
- **Flags (3 bits):** Controls fragmentation. The bits indicate if the packet can be fragmented and if there are more fragments to follow.
- **Fragment Offset (13 bits):** Indicates where in the original datagram a particular fragment belongs.
- **Time to Live (TTL) (8 bits):** A hop counter that prevents datagrams from looping indefinitely. Each router decrements this value by one; if it reaches zero, the datagram is discarded.
- **Protocol (8 bits):** Identifies the next-level protocol (e.g., TCP = 6, UDP = 17) to which the data should be passed.
- **Header Checksum (16 bits):** Used for error checking of the header only. It is recalculated at every hop because the TTL field changes.
- **Source IP Address (32 bits):** The 32-bit IP address of the sender.
- **Destination IP Address (32 bits):** The 32-bit IP address of the intended recipient.
- **Options (Variable):** Optional fields for testing, debugging, or security. If used, padding is added to ensure the header ends on a 32-bit boundary.

2.4- Classful IPv4 Addressing

IPv4 addressing, at its inception, used the concept of classes. This architecture is called classful addressing. Although this scheme is becoming obsolete, we briefly discuss it here to show the rationale behind classless addressing.

2.4.1- Classes

In classful addressing, the address space is divided into **five classes: A, B, C, D, and E**. Each class occupies some part of the address space.



2.4.2- Private IP addresses

These addresses can be used on a private network, but they are not routable through the internet. The range for the private IP addresses is given in table below

Range	Prefix
10.0.0.0 – 10.255.255.255	/ 8
172.16.0.0 – 172.31.255.255	/ 12
192.168.0.0 – 192.168.255.255	/ 16

The last column of Table above shows the mask in the form (/n) where n can be 8, 16, or 24 in classful addressing. This notation is also called *prefix notation* or **Classless Interdomain Routing (CIDR) notation**. The notation is used in classless addressing. We introduce it here because it can also be applied to classful addressing.

2.5- Classless Addressing

Classful allocation of address space often wasted many addresses, which exhausted the availability of IPv4 addresses. For example, a company that had a network with 260 hosts would need to be given a class B address with more than 65,000 addresses. The system that we currently use is referred to as classless addressing. With the classless system, address blocks appropriate to the number of hosts are assigned to companies or organizations without regard to the unicast class. This is can be achieved using many techniques such as:

- Subnetting
- Variable Length Subnet Mask (VLSM)
- Network Address Translation (NAT)

2.5.1- Subnetting

For the purpose of *reducing network traffics* and *simplifying network management*, and also *reducing the waste in IP addresses*; large network can be divided into smaller networks (subnetworks). Subnetting allows for creating multiple logical networks from a single address block.

We create the subnets by using one or more of the host bits as network bits. *This is done by extending the mask to borrow some of the bits from the host portion of the address to create additional network bits.* The more host bits used, the more subnets that can be defined. For each bit borrowed, we double the number of subnetworks available.

Notes:

- *The maximum number of borrowed bits is by leaving 2-bits in the HostID part of the address.*
- To determine the number of subnets or hosts use the following rules:

$$\text{Number of Subnets} = 2^{\text{number_of_borrowed_host_bits}}$$

$$\text{Usable Hosts Per Subnet} = (2^{\text{number_of_remaining_host_bits}} - 2)$$

Example: What are the number of subnets, hosts, and the valid host addresses given class C IP address 192.168.10.5 and a subnet mask 255.255.255.192 (192.168.10.5 / 26)

Solution:

IP address → 192.168.10.5 → 11000000 . 10101000 . 00001010 . 00000101

Subnet mask → 255.255.255.192 → 11111111 . 11111111 . 11111111 . 11000000

Network address → 192.168.10.0 → 11000000 . 10101000 . 00001010 . 00000000

Subnet mask → 255.255.255.192 → 11111111 . 11111111 . 11111111 . 11⁽⁰⁾000000

2 bits are borrowed

The number of subnets = 2^n . (where n = the number of bits borrowed)

$$= 2^2 = 4 \text{ subnets}$$

The number of hosts on each subnet = $2^m - 2$. (where m = the number of bits left for hosts).

$$= 2^6 - 2 = 62$$

shows that each of these subnets can have 62 hosts.

For each subnet, examine the last octet in binary. The values in these octets for the four networks are:

Subnet 1: **00000000** = 0

Subnet 2: **01000000** = 64

Subnet 3: **10000000** = 128

Subnet 4: **11000000** = 192

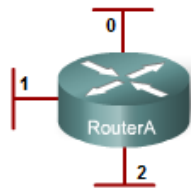
Or the subnet block size is = $2^{\text{number-of-host-bits}} = 2^6 = 64$ (*the magic number*)

Therefore, the subnets are = (0, 64, 128, 192)

Table below shows the valid host addresses and the subnet and broadcast addresses for each subnet.

Subnet	Subnet address	Host range	Broadcast address
1	192.168.10.0	192.168.10.1 → 192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65 → 192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129 → 192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193 → 192.168.10.254	192.168.10.255

Figure below illustrate the same process



-	192.168.1.0 (/24)	Address:	11000000.10101000.00000001.00000000
	255.255.255.0	Mask:	11111111.11111111.11111111.00000000
0	192.168.1.0 (/26)	Address:	11000000.10101000.00000001.00000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000
1	192.168.1.64 (/26)	Address:	11000000.10101000.00000001.01000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000
2	192.168.1.128 (/26)	Address:	11000000.10101000.00000001.10000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000
3	192.168.1.192 (/26)	Address:	11000000.10101000.00000001.11000000
	255.255.255.192	Mask:	11111111.11111111.11111111.11000000

Two bits are borrowed to provide four subnets.

Unused address in this example.

A 1 in these positions in the mask means that these values are part of the network address.

More subnets are available, but fewer addresses are available per subnet.

2.5.2- VLSM

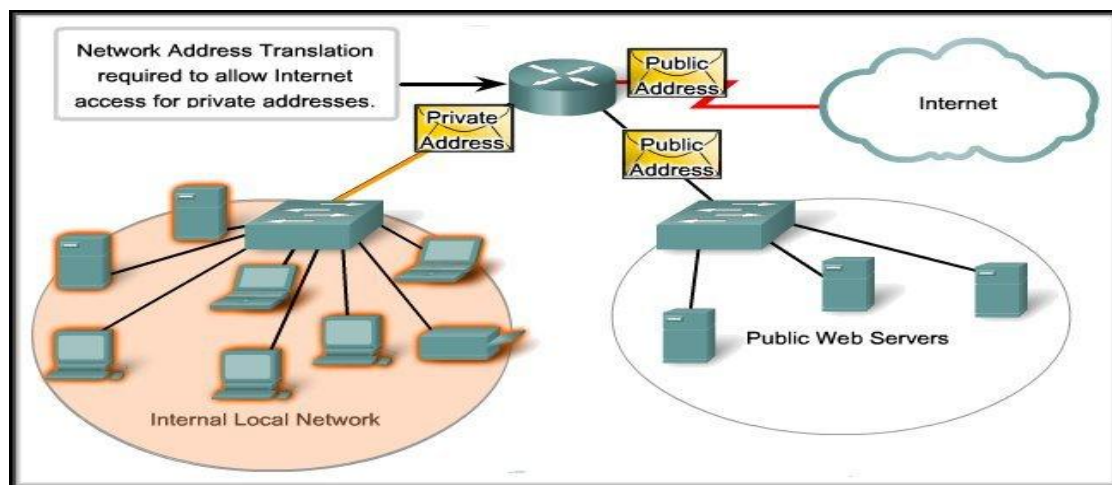
A serious limitation of using only a single subnet mask across a given network was that an organization is locked into a fixed-number of fixed-sized subnets. The solution to this problem is to use a **VLSM (variable length subnet mask)** enables a network number to be configured with different subnet masks on different interfaces.

2.5.3- Network Address Translation (NAT)

The number of home users and small businesses that want to use the Internet is ever increasing. The distribution of addresses through ISPs has created a new problem. If the business grows or the household needs a larger range, the ISP may not be able to grant the demand because the addresses before and after the range may have already been allocated to other networks. In most situations, however, only a portion of computers in a small network need access to the Internet simultaneously.

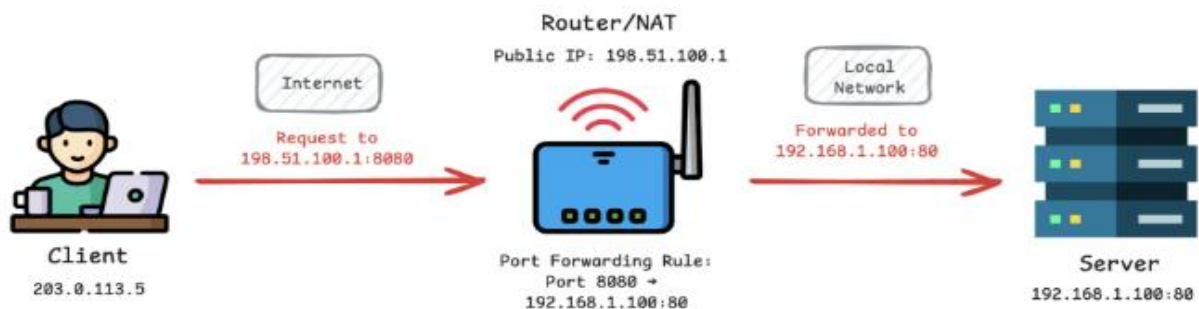
A quick solution to this problem is called **network address translation (NAT)**. NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally. The traffic inside can use the large set; the traffic outside,

the small set. The set of **private addresses** can be used **inside** the home or business network as shown in figure below.



2.5.4- Port Forwarding

Port Forwarding or **Port Mapping** is an application of network address translation (NAT) that redirects a communication request from one **address and port number combination** to another while the packets are traversing a network gateway, such as a router or firewall. This technique is most commonly used to make services on a host residing on a **protected or masqueraded** (internal) network available to hosts on the opposite side of the gateway (external network), by remapping the destination IP address and port number of the communication to an internal host

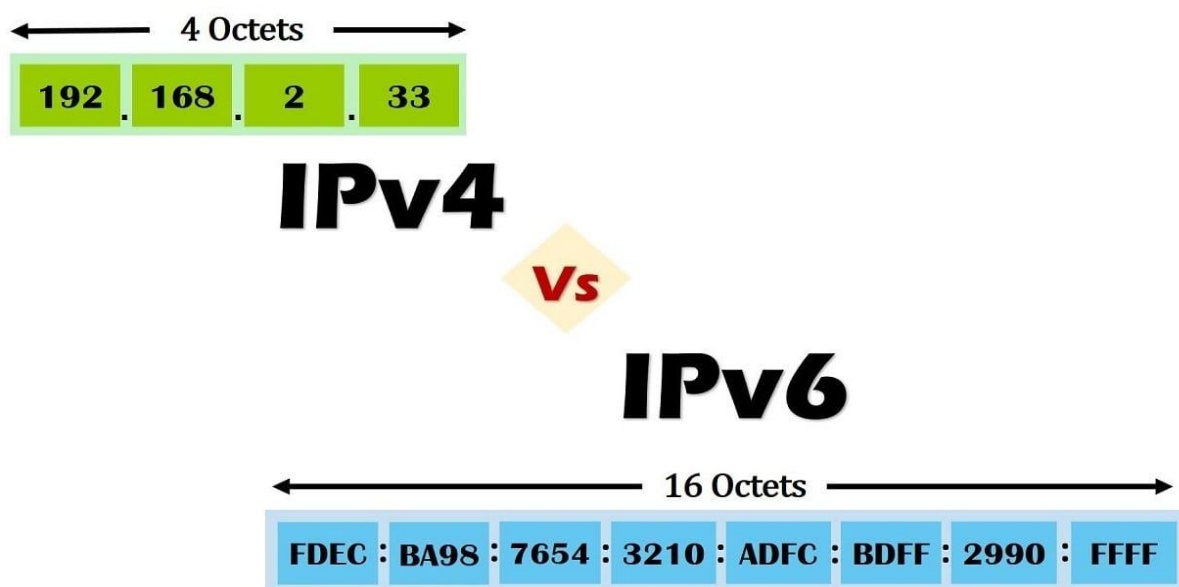


1. Client sends request to public IP (198.51.100.1) on port 8080
2. Router receives the request and checks port forwarding rules
3. Router forwards the request to internal server (192.168.1.100) on port 80
4. Server processes the request and sends response back through the same path

2.6- IPv6 Addressing

The need for more addresses, in addition to other concerns about the IP layer, motivated a new design of the IP layer called the new generation of IP or IPv6 (IP version 6). In this version, the Internet uses **128-bit addresses** that give much greater flexibility in address allocation. These addresses are referred to as IPv6 (IP version 6) addresses.

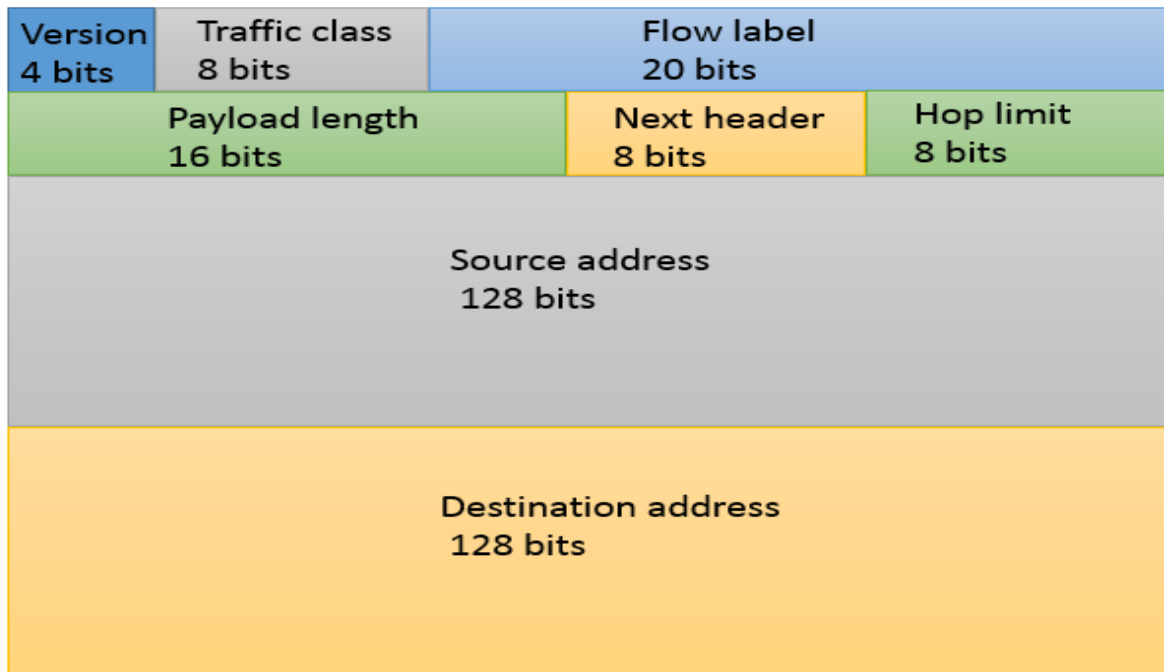
An IPv6 address consists of 16 bytes (octets); it is 128 bits long. To make addresses more readable, IPv6 specifies hexadecimal colon notation. In this notation, 128 bits is divided into eight sections, each 2 bytes in length. Two bytes in hexadecimal notation requires four hexadecimal digits. Therefore, the address consists of 32 hexadecimal digits, with every four digits separated by a colon, as shown below



2.6.1- IPv6 Features

The ability to scale networks for future demands requires a limitless supply of IP addresses and improved mobility; IPv6 combines expanded addressing with a more efficient and feature-rich header to meet these demands.

- header format helps speed processing/forwarding
- header changes to facilitate QoS



The main benefits of IPv6 include the following:

✚ **Larger address space:**

IPv6 addresses are 128 bits, this larger addressing space allows more support for addressing hierarchy levels, a much greater number of addressable nodes, and a simpler auto-configuration of addresses.

✚ **Globally unique IP addresses:**

Every node can have a unique global IPv6 address, which eliminates the need for NAT.

✚ **Header format efficiency:**

A simplified header with a fixed header size makes processing more efficient.

✚ **Improved privacy and security:**

IPsec is standard for IP network security, available for both IPv4 and IPv6.

✚ **Multicast instead of broadcast:**

With IPv6 the multicast communication substitutes the needs of broadcasting by using all node multicast

✚ **Stateless and stateful address configuration:**

IPv6 support besides the stateful addresses (DHCP) a stateless addresses autoconfiguration feature (SLAAC)

2.6.2- Compressed IPv6 Representation

There are two rules used to compress the IPv6 Representation

- **Rule 1 (Leading Zero Compression):** Eliminate the starting zero(s) from any hextex (zeros to the left).
- **Rule 2 (Successive Zeros Compression):** Eliminate group of successive zeros by one double-colon (::) but you can perform this only once in your IPv6 address.

Example: Given the following IPv6 address

(2001:1265:0000:0000:0AE4:0000:005B:06B0) used the rules to re-represent it in a compressed format.

Solution:

Original IP: 2001:1265:0000:0000:0AE4:0000:005B:06B0

After using Rule 1 the IP can be compressed into: 2001:1265:0:0:AE4:0:5B:6B0

After using Rule 2 the IP can be reduced into: 2001:1265::AE4:0:5B:6B0

2.6.3- IPv6 Three Address Types

In IPv6, there are three categories of IPv6 addresses:

- **Unicast**
- **Multicast**
- **Anycast**

Unicast Address:

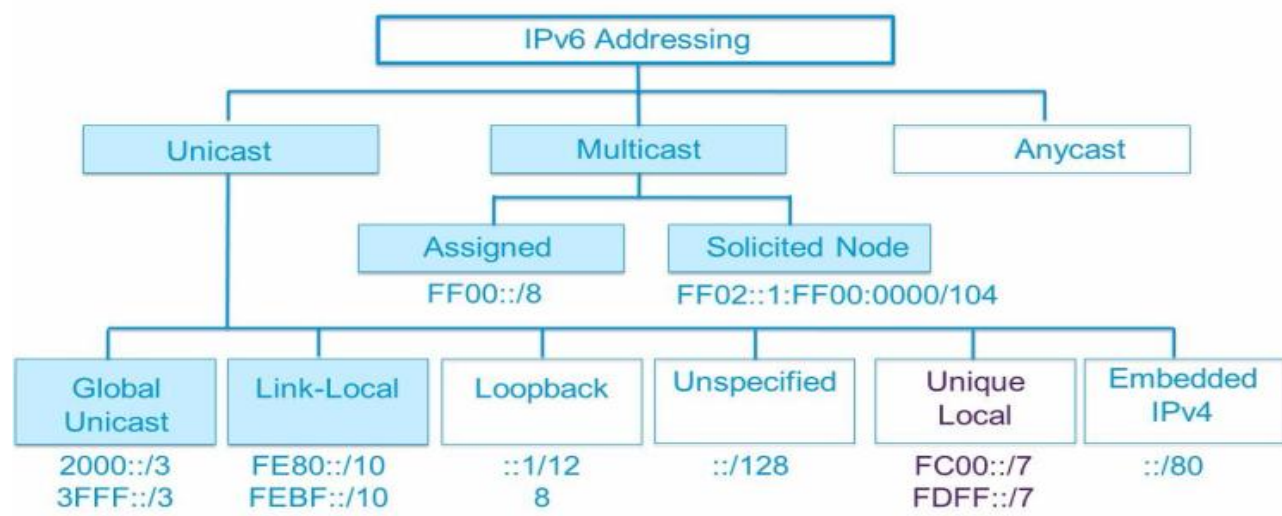
A unicast address defines a single interface (computer or router). The packet sent to a unicast address will be routed to the intended recipient.

Multicast Address:

A multicast address also defines a group of computers. In multicasting, each member of the group receives a copy.

Anycast Address:

An anycast address defines a group of computers that all share a single address. A packet with an anycast address is delivered to only one member of the group, the most reachable one.

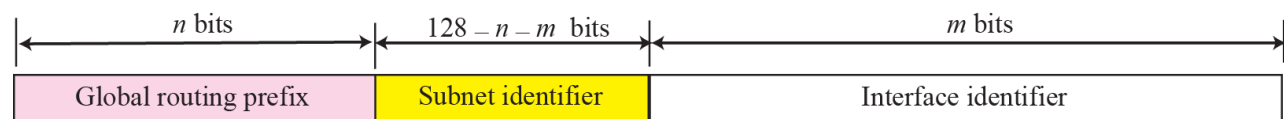


2.6.4- Three Levels of Hierarchy

An address in this block is divided into three parts:

- Global routing prefix
- Subnet identifier
- Interface identifier

as shown in Figure below:

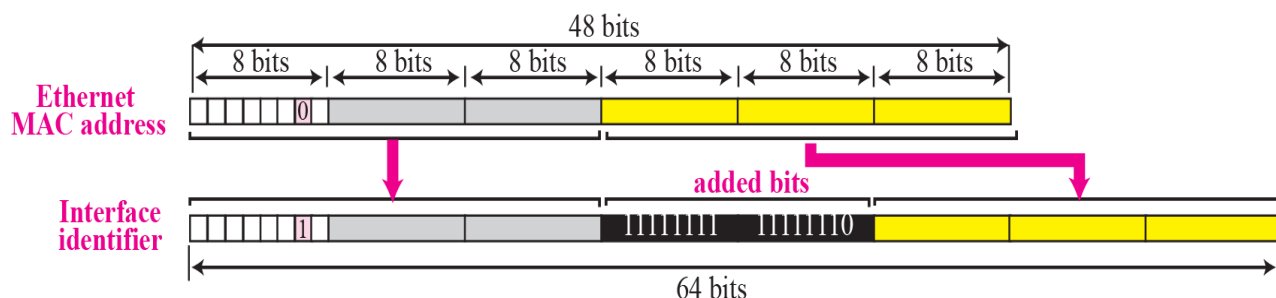


2.6.5- IPv6 Autoconfiguration

One of the interesting features of IPv6 addressing is the **autoconfiguration** of hosts. As we discussed in IPv4, the host and routers can be configured manually by the network manager. However, the Dynamic Host Configuration Protocol, DHCP, can be used to allocate an IPv4 address to a host that joins the network. In IPv6, DHCP protocol can still be used to allocate an IPv6 address to a host, but a host can also configure itself.

Interface Identifier

- The last 64 bits define the interface identifier.
- The interface identifier is similar to hostid in IPv4 addressing.
- The IPv6 addressing allows this opportunity. A physical address whose length is less than 64 bits can be embedded as the whole or part of the interface identifier, eliminating the mapping process. Mapping the 48-bit physical address defined by Ethernet to 64-bit extended unique identifier (EUI-64).



Example: Find the interface identifier if the Ethernet physical address is **(F5-A9-23-14-7A-D2)₁₆** using the format we defined for Ethernet addresses.

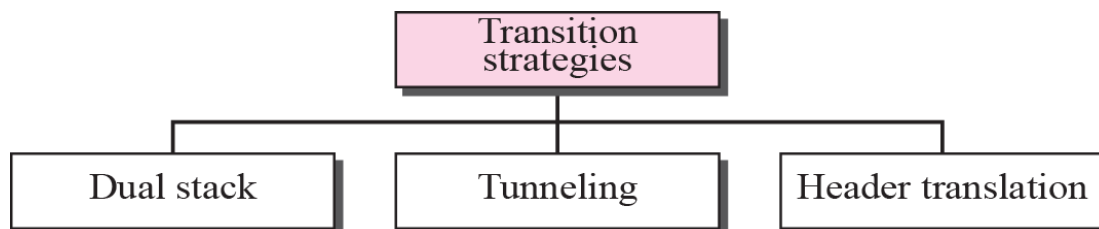
Solution:

We only need to change the seventh bit of the first octet from 0 to 1, insert two octets **(FFFE)₁₆** and change the format to colon hex notation.

The result is **F7A9:23FF:FE14:7AD2** in colon hex.

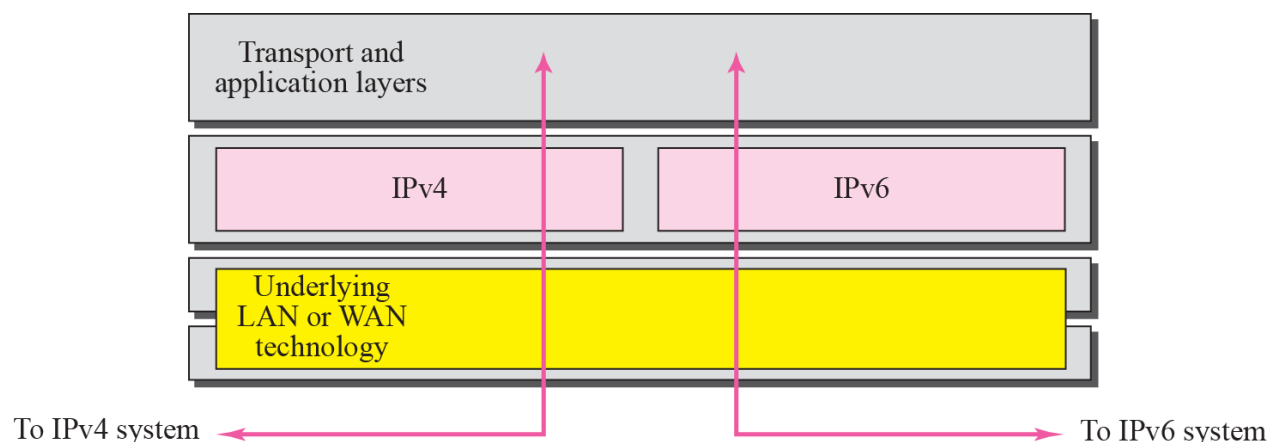
2.7- Transition from IPv4 to IPv6

Because of the huge number of systems on the Internet, the transition from IPv4 to IPv6 cannot happen suddenly. It will take a considerable amount of time before every system on the Internet can move from IPv4 to IPv6. The transition must be smooth to prevent any problems between IPv4 and IPv6 systems. Three strategies have been devised by the IETF to help the transition.



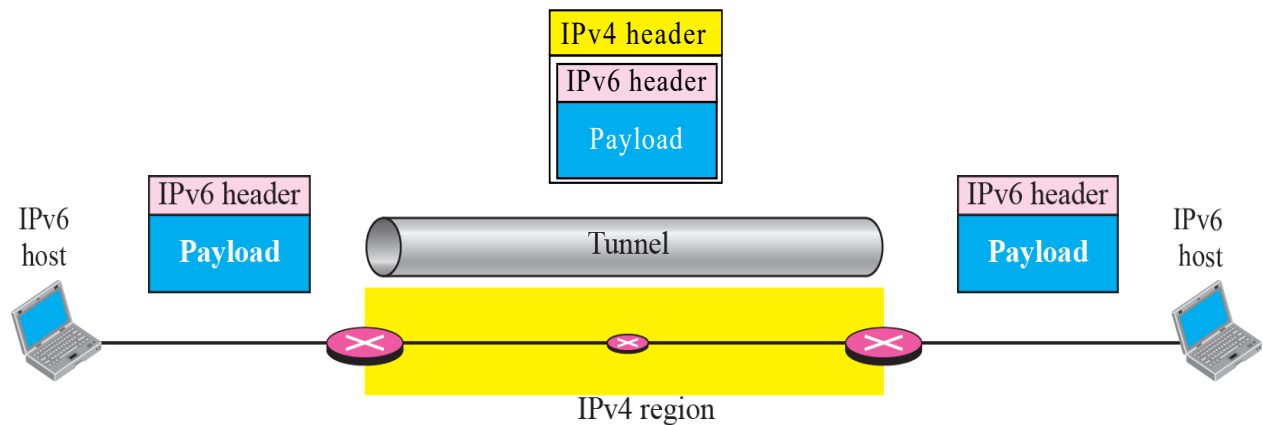
Dual Stack

It is recommended that all hosts, before migrating completely to version 6, have a dual stack of protocols. In other words, a station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6.



Tunneling

Tunneling is a strategy used when two computers using IPv6 want to communicate with each other, and the packet must pass through a region that uses IPv4. To make it clear that the IPv4 packet is carrying an IPv6 packet as data



Header Translation

Header translation is necessary when the majority of the Internet has moved to IPv6, but some systems still use IPv4. The sender wants to use IPv6, but the receiver does not understand IPv6. Tunneling does not work in this situation because the packet must be in the IPv4 format to be understood by the receiver. In this case, the header format must be totally changed through header translation (NAT64). The header of the IPv6 packet is converted to an IPv4 header.

